Predicting Financial Market Stress Regimes Using Multi-Asset Machine Learning

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***Abstract*—Early detection of systematic market stress remains a central challenge in quantitative finance, as traditional risk models often rely on lagging indicators that fail to capture sudden cross-asset panic. This paper introduces a cross-asset machine learning model classifying trading days into three distinct states (Low, Normal, and High Stress) using short-term momentum velocity, rolling volatility across equities, and fixed income. Using historical data from 2011 to 2025, we evaluate a non-linear Random Forest classifier against a linear Logistic Regression baseline across a strict chronological test/train split. The Random Forest Model achieved an out-of-sample accuracy of 59.21%, outperforming the Logistic Regression baseline by 18.52 percentage points. Our feature importance results demonstrate that equity momentum velocity acts as a critical leading indicator of market distress offering quantitative risk managers an earlier warning system before liquidity shock spreads to slower moving assets.**

***Index Terms*—Machine Learning, Random Forest, Feature Engineering, Market Stress Detection, Cross-Asset Dynamics, Quantitative Finance.**

I. Introduction

Market stress is defined as periods where financial markets undergo high volatility and increased uncertainty [1]. Whether this stress is caused by geopolitical events, investor psychology, or financial crises, its effects can range from investors fleeing to safe-haven assets to systemic economic fallout like mass unemployment. Early forecasting of these stress periods is therefore critical, as it allows investors and institutions to adjust their risk management strategies before losses accumulate.

The CBOE Volatility Index (VIX) measures the market's expectation of 30-day volatility for the S&P 500. Popularly known as "the fear gauge," the VIX spikes when investors are nervous and drops when they are confident [2]. The VIX has proven highly responsive during major modern disruptions, including the 2020 COVID-19 market crash, the 2022 Federal Reserve monetary tightening cycle, and the 2025 market volatility spikes, all of which are captured in our study's time period. Because of its demonstrated sensitivity to shifts in market sentiment, the VIX serves as an ideal metric for defining and labeling market stress regimes in this study.

Practitioners and researchers historically rely on traditional methods such as financial stress indices, static volatility thresholds, and single-asset indicators to evaluate market conditions. However, these traditional measures typically look at a single asset or index in isolation, thereby failing to capture how stress spreads across bonds, equities, and commodities simultaneously. Machine learning offers a more promising approach, as it can identify complex, non-linear relationships across multiple distinct asset classes at the same time to forecast shifting conditions [3].

This paper proposes a machine learning system to predict whether a future trading day will fall into a Low, Normal, or High stress regime. Our data consists of U.S. equities (SPY and QQQ), fixed income (BND), and commodities (GLD), spanning 3,768 trading days from April 2011 to September 2025. From this data, target stress regimes were labeled using VIX percentiles, where days below the 33rd percentile were classified as Low stress, between the 33rd and 66th percentiles as Normal, and above the 66th percentile as High stress. To shift the system from a passive detector to an active predictor, we engineered trend-based features including 10 and 20-day momentum, drawdowns, short-term lagged returns, and rolling volatility for each asset.

We benchmarked two distinct classification models on this predictive task. Our baseline model, Logistic Regression, is a transparent linear architecture, whereas our primary classifier, Random Forest, leverages an ensemble of decision trees to map complex, non-linear patterns. Beyond simply predicting stress regimes, this paper aims to identify which specific historical features and assets contribute most to forward-looking market instability, offering actionable insights for risk management practitioners.

Tested on entirely unseen out-of-sample data spanning 2022 to 2025, our Random Forest classifier achieved a highly competitive 59.21% accuracy, outperforming the Logistic Regression baseline (40.69%) by 18.52 percentage points. Feature importance metrics revealed that medium-term equity velocity indicators, specifically NASDAQ 20-day momentum and S&P 500 10-day momentum, were the dominant predictors of future market stress. The remainder of this paper is structured as follows: Section 2 outlines the feature engineering, methodology, and model selection; Section 3 presents the empirical results; Section 4 discusses the findings and limitations; and Section 5 concludes.

II. Data and Methodology

Financial markets are composed of different asset classes that respond differently to stress periods; therefore, capturing an accurate understanding of market stress requires a multi-asset analysis. This study focuses on four major assets: the SPDR S&P 500 ETF (SPY) for broad U.S. equities, the Invesco QQQ Trust (QQQ) for technology equities, the Vanguard Total Bond Market ETF (BND) for representing the U.S. fixed-income market, and the SPDR Gold Shares ETF (GLD) as a commodity and safe-haven asset. This diversity of

asset classes allows the study to capture how stress manifests differently across equities, bonds, and commodities.

A. Data Collection and Preprocessing

This study's data was sourced from Yahoo Finance, spanning from April 2011 to September 2025 and encompassing 3,768 trading days. Days on which federal markets were closed were systematically excluded from the dataset. We used Adjusted Close prices as opposed to raw closing prices, since adjusted close prices account for dividends and stock splits, ensuring historical price continuity.

The selected time period was chosen to capture a diverse range of market conditions. Between 2011 and 2025, markets underwent major bull markets, sharp technical corrections, and systemic crises, including the 2020 COVID-19 market crash, the 2022 Federal Reserve monetary tightening cycle, and the 2023–2025 recovery period. Such variety ensures the model is trained across different stress environments rather than a single market regime, which in turn strengthens the out-of-sample generalizability of our findings.

B. Feature Engineering

To ensure the models actively forecast forward-looking market states rather than merely describing contemporaneous conditions, raw price series were transformed into trend-based features. Instead of raw asset prices, we computed 10-day and 20-day momentum, rolling historical volatility, and lagged returns across horizons of 1, 2, 3, and 5 days.

To achieve stationarity and eliminate level dependencies across asset prices, we applied a daily first-difference transformation to these engineered features:

$$\Delta X_t = X_t - X_{t-1} \tag{1}$$

where X_t denotes the feature value on trading day t and X_{t-1} denotes the corresponding value on the preceding trading day. Because this operation induces a single missing observation at $t = 1$, the initial row was removed to maintain dataset integrity. This transformation prevents models from anchoring to absolute price levels and forces them to learn from day-over-day velocity and rate-of-change dynamics.

To capture expanding market risk and localized variance, rolling volatility was computed for each asset. Daily logarithmic returns, P_t, were first computed from adjusted closing prices:

$$P_t = \ln\left(\frac{X_t}{X_{t-1}}\right) \tag{2}$$

Using daily log returns, the rolling sample standard deviation σ_t was calculated over rolling windows:

$$\sigma_t = \sqrt{\frac{1}{n-1}\sum_{i=0}^{n-1}\left(P_{t-i} - \bar{P}\right)^2} \tag{3}$$

where:

- P_t is the logarithmic return on trading day t.
- σ_t represents the localized rolling volatility on day t.
- $n \in \{10, 20\}$ denotes the rolling sample window length.
- P_{t-i} is the log return i trading days prior to day t.
- $\bar{P} = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}$ represents the sample mean of log returns across the active window.

C. Target Variable and Train-Test Partitioning

Our target variable, designated as `Stress_levels_ALL`, categorizes market environments into three discrete regimes: Low, Normal, and High stress.

To evaluate predictive performance in an authentic out-of-sample setting, we implemented a strict chronological split rather than random sampling:

- **Training Set:** Historical observations from April 2011 through December 31, 2021.
- **Test Set:** Out-of-sample observations from January 1, 2022, through September 2025 (1,015 trading days).

Chronological partitioning eliminates look-ahead bias and tests the model's resilience across major macroeconomic regime shifts.

D. Model Selection and Normalization

We evaluated two distinct architectures on the multi-class regime prediction task:

- **Logistic Regression:** Serves as a transparent linear baseline to evaluate linear separability among stress regimes.
- **Random Forest Classifier:** Deployed to capture complex non-linear interactions, such as sudden momentum decelerations coinciding with localized volatility spikes.

To prevent data leakage, feature scaling was conducted using `StandardScaler` [4], with transformation parameters fitted strictly on the training partition and subsequently applied to the out-of-sample test partition:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{4}$$

where μ_{train} and σ_{train} represent the empirical mean and standard deviation of the training set features, respectively.

To capture rate-of-change dynamics and velocity across asset classes, rolling price momentum was computed over a defined historical window:

$$M_{t,n} = X_t - X_{t-n} \tag{5}$$

where:

- $M_{t,n}$ denotes the rolling price momentum on trading day t over lookback window n.
- X_t denotes the adjusted closing price on day t.
- X_{t-n} denotes the adjusted closing price n trading days prior to day t.
- $n \in \{10, 20\}$ denotes the lookback interval in trading days.

To capture cross-asset contagion effects, engineered features from equities (SPY, QQQ), fixed income (BND), and safe-haven commodities (GLD) were concatenated into a unified daily observation vector:

$$\mathbf{M}_t = \begin{bmatrix}\mathbf{f}_{\text{SPY},t} & \mathbf{f}_{\text{QQQ},t} & \mathbf{f}_{\text{BND},t} & \mathbf{f}_{\text{GLD},t}\end{bmatrix} \tag{6}$$

where $\mathbf{f}_{a,t}$ denotes the composite vector of momentum, lagged returns, and volatility features for asset a on trading day t.

To ensure numerical stability against extreme outliers, features were winsorized by capping values outside the 1st and 99th percentiles based strictly on the training partition. All time-series data was partitioned chronologically into training (70%) and out-of-sample testing (30%) sets to prevent look-ahead bias and data leakage.

III. Methodology and Model Architecture

A. Classification Models and Non-Linearity

To assess multi-asset market stress, we compare the predictive performance of an ensemble non-linear Random Forest classifier against a generalized linear Logistic Regression baseline [5]. Financial crises, liquidity shocks, and regime transitions represent intrinsically non-linear market phenomena. While standard linear models assume linear separability across the feature space, tree-based ensemble architectures recursively partition the feature domain into lower-dimensional subspaces to approximate complex conditional probability distributions.

The Random Forest model aggregates predictions across an ensemble of B randomized decision trees [6]:

$$\hat{y} = \text{mode}\left\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_B(\mathbf{x})\right\} \tag{7}$$

where:

- $\hat{y}$ denotes the final ensemble predicted stress regime.
- $T_b(\mathbf{x})$ represents the individual classification output of decision tree b for feature vector $\mathbf{x}$.
- B is the total number of bootstrap-aggregated estimators ($B = 100$).

By training each individual tree on a bootstrap sample with randomized feature subsets at each node, bootstrap aggregation (bagging) reduces model variance and captures complex multi-asset interactions, such as sharp equity momentum deceleration during periods of elevated volatility.

B. Target Variable Definition

We define the target classification variable $y_t \in \{0, 1, 2\}$ as a function of multi-asset drawdowns and historical volatility regimes:

$$y_t = \begin{cases} 0 & \text{Class 0: Low Stress} \\ 1 & \text{Class 1: Normal Stress} \\ 2 & \text{Class 2: High Stress} \end{cases} \tag{8}$$

C. Model Configuration and Class Weighting

Given the inherent class imbalance of financial stress regimes—where crisis periods occur far less frequently than normal baseline market conditions—custom penalty loss weights were applied to prevent majority-class bias and algorithmic degeneracy.

1) Random Forest Classifier:

- **Number of Estimators (B):** 100 decision trees.
- **Splitting Criterion:** Gini Impurity (I_G), defined across $C = 3$ stress regimes as:

$$I_G(p) = 1 - \sum_{i=1}^{C} p_i^2 \tag{9}$$

  where p_i is the empirical probability of an observation being classified into class i at a given node.
- **Class Weight Penalties:** High Stress = 1.0, Normal Stress = 2.0, Low Stress = 15.0.

2) Logistic Regression Baseline:

- **Weight Balancing:** `class_weight='balanced'` (weights adjusted dynamically in inverse proportion to class frequencies).
- **Maximum Iterations:** 1,000 iterations to guarantee numerical convergence.

D. Chronological Split and Leakage Prevention

Standard randomized train-test splits introduce look-ahead bias and temporal leakage when applied to time-series data. To preserve temporal causality, datasets were partitioned chronologically at December 31, 2021:

- **Training Set:** April 2011 through December 31, 2021.
- **Out-of-Sample Test Set:** January 1, 2022 through September 9, 2025.

To prevent information leakage from future market conditions into trained parameters, first-order differencing was applied strictly using backward-looking windows. Feature normalization was performed via `StandardScaler`, fitted exclusively on the training partition and applied out-of-sample to the test partition.

IV. Empirical Results

This section presents the empirical evaluation of the Random Forest ensemble classifier against the Logistic Regression baseline. Performance is evaluated across global classification metrics, regime-specific precision-recall dynamics, and Gini feature importance contributions.

A. Macro Performance Comparison

Both models were evaluated out-of-sample on identical test partitions spanning January 1, 2022, to September 9, 2025 ($N = 1{,}015$ trading days). Prior to evaluation, all input series were stationarized via first-differencing and normalized using parameters fitted solely on the training partition.

As detailed in Table I, the Random Forest classifier achieved a global accuracy of 59.21%, outperforming the linear baseline by 18.52 percentage points. The ensemble model demonstrated consistent performance gains across all aggregate statistical metrics, yielding a 16.00 percentage point increase in macro precision and a 12.00 percentage point improvement in macro F1-score.

TABLE I
MACRO PERFORMANCE COMPARISON ON OUT-OF-SAMPLE TEST SET (2022–2025)

Classification Model	Global Accuracy	Macro Precision	Macro Recall	Macro F1-Score
Logistic Regression (Baseline)	40.69%	41.00%	41.00%	39.00%
Random Forest (Ensemble)	**59.21%**	**57.00%**	**52.00%**	**51.00%**

B. Performance Breakdown by Stress Regime

1) Logistic Regression Baseline: To prevent majority-class collapse, the baseline model was configured with synthetic class balancing (`class_weight='balanced'`). While this optimization forced the linear boundary to actively allocate classifications across all three regimes (achieving a macro recall of 41.00%), its linear functional form was unable to resolve the non-linear boundaries separating multi-asset stress phases, resulting in a low macro F1-score of 39.00%.

2) Random Forest Ensemble: The Random Forest model effectively mapped the conditional boundaries of the target space. Under asymmetric class weights ($w_{\text{High}} = 1.0, w_{\text{Normal}} = 2.0, w_{\text{Low}} = 15.0$), the model demonstrated exceptional performance in identifying crisis states:

- **High Stress Regime:** Achieved 79.00% precision and 68.00% recall. When generating a high-stress warning, the model exhibits high predictive reliability.
- **Low Stress Regime:** Achieved 46.00% precision and 15.00% recall. Performance attenuation in this class reflects the structural scarcity of low-volatility days during the post-2022 monetary tightening cycle (comprising only 203 of the 1,015 test observations).

C. Feature Importance Analysis

Mean Decrease in Impurity (MDI) was extracted from the trained Random Forest ensemble to quantify the predictive contribution of each engineered signal. Because features represent stationary first-differences rather than nominal levels, these rankings isolate the dynamical properties that drive regime shifts.

TABLE II
TOP PREDICTIVE FEATURES RANKED BY GINI FEATURE IMPORTANCE

Rank	Feature Description	Asset Class	MDI Importance
1	NASDAQ 20-Day Momentum	Equities (QQQ)	0.0751
2	S&P 500 10-Day Momentum	Equities (SPY)	0.0729
3	NASDAQ Rolling Volatility	Equities (QQQ)	0.0508
4	S&P 500 Drawdown	Equities (SPY)	0.0389
5	S&P 500 Rolling Volatility	Equities (SPY)	0.0387

The empirical rankings in Table II demonstrate that intermediate-horizon equity momentum and localized equity variance are the primary drivers of regime classification. Specifically, the heavy weighting of NASDAQ 20-day momentum (0.0751) and S&P 500 10-day momentum (0.0729) indicates that multi-week equity velocity carries substantially more forward-looking predictive signal than isolated single-day price shocks or fixed-income metrics.

V. DISCUSSION

A. Linearity vs. Non-Linearity in Regime Detection

The empirical results demonstrate a substantial performance advantage for the Random Forest classifier (59.21% accuracy) over the Logistic Regression baseline (40.69% accuracy). The 18.52 percentage point spread highlights the architectural limitations of classical linear modeling when evaluating systemic financial risk. Standard linear classifiers assume constant marginal effects and feature independence. Under a linear specification, a 1.00% drawdown in equity indices is assigned an invariant risk weight regardless of prevailing macroeconomic conditions.

In financial markets, however, regime transitions are intrinsically state-dependent and governed by non-linear thresholds. A localized decline in price momentum during an economic expansion may produce negligible market stress, whereas an identical decline during a high-volatility regime can trigger widespread liquidity evaporation and cascade into systemic panic. By recursively partitioning the feature space across an ensemble of decision trees, the Random Forest architecture successfully models these non-linear conditional interactions without requiring rigid functional parametric assumptions.

B. Equity Velocity as a Leading Stress Indicator

Gini feature importance rankings revealed that intermediate-horizon equity velocity—specifically NASDAQ 20-day momentum and S&P 500 10-day momentum—dominated model decision boundaries. Traditional macro-prudential frameworks frequently rely on lagging or contemporaneous risk measures, such as maximum drawdowns and trailing realized variances. Our empirical findings indicate that rate-of-change indicators provide superior forward-looking warning signals.

Because large-cap equities and tech-heavy indices (SPY, QQQ) serve as primary liquidity vehicles for institutional portfolio rebalancing, sharp decelerations in medium-term price velocity reflect early-stage institutional de-risking and capital flight. This structural deleveraging registers in equity momentum dynamics significantly earlier than stress manifests in slower-moving credit, aggregate fixed-income, or commodity series.

C. Macro-Structural Regime Shifts and Distributional Drift

The primary performance bottleneck—characterized by attenuated precision (46.00%) and recall (15.00%) for the "Low Stress" class—stems from a macro-structural distribution shift rather than algorithmic mis-specification. The training dataset (2011–2021) captured an era of secular stagnation characterized by Zero Lower Bound (ZLB) interest rates, quantitative

easing, and depressed baseline volatility. Conversely, the out-of-sample test window (2022–2025) was marked by rapid Federal Reserve monetary tightening, persistent inflation shocks, and elevated geopolitical friction.

This structural break shifted baseline volatility upward across all asset classes. Under the post-2022 macro environment, a relatively calm trading session exhibited statistical characteristics comparable to a moderate stress period in the pre-2021 training distribution. Consequently, historical low-stress archetypes were both structurally scarce (comprising only 20.00% of the test set) and difficult for the model to isolate without suffering false-positive contamination.

D. Limitations and Future Research Directions

While our multi-asset framework captures cross-market volatility spillovers across equities, fixed income, and commodities, several avenues remain for future research:

- **Macroeconomic Event Integration:** The current model relies exclusively on technical, end-of-day price series. Incorporating scheduled macroeconomic releases—such as Federal Open Market Committee (FOMC) rate decisions, Consumer Price Index (CPI) inflation prints, and non-farm payroll reports—could enhance predictive precision prior to volatility spikes.
- **Sequential and Gradient-Boosted Architectures:** Future iterations could benchmark Random Forests against recurrent neural networks (e.g., Long Short-Term Memory networks) [7] to capture long-range temporal dependencies, as well as gradient-boosted decision trees (XGBoost, LightGBM) to refine classification margins.
- **Dynamic Lookback Calibration:** Testing adaptive rolling calibration windows instead of a fixed historical training partition could help the model adapt dynamically to shifting monetary regimes and baseline volatility drifts.

VI. Conclusion

This study evaluated the efficacy of non-linear machine learning architectures in forecasting multi-asset market stress regimes. By constructing an engineered pipeline capturing cross-asset momentum, rolling volatility, and lagged returns across equities, fixed income, and commodities, our Random Forest classifier achieved an out-of-sample accuracy of 59.21%, outperforming the linear Logistic Regression baseline (40.69%) by 18.52 percentage points.

While conventional risk management paradigms rely heavily on contemporaneous or lagging measures such as trailing drawdowns, our empirical findings demonstrate that price velocity and momentum serve as critical early warning indicators of market distress. Given the liquidity profile of large-cap and technology equities, institutional investors execute initial de-risking maneuvers in equity markets before reallocating across less liquid instruments. Consequently, sharp decelerations in equity momentum register days or weeks before systemic pressure manifests in fixed-income yields or commodity curves.

Finally, out-of-sample precision trade-offs underscore the challenges of macroeconomic distribution shifts. The training partition (2011–2021) reflected a prolonged low-rate, low-volatility monetary regime, whereas the testing window (2022–2025) introduced aggressive Federal Reserve monetary tightening, persistent inflation, and elevated baseline variance. This macroeconomic evolution altered the statistical profile of baseline market conditions, rendering low-stress regimes structurally difficult to isolate. Future research can build upon this multi-asset framework by integrating macroeconomic data releases (e.g., CPI prints, FOMC rate decisions) and deploying sequential deep learning models such as LSTMs or gradient-boosted architectures.

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